# Hints: Integrals – 5. The gamma and beta functions and Wallis' formulas

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

---

## Problem 1

**Hint 1.** Write $\Gamma(p)\Gamma(q) = \int_0^\infty \int_0^\infty x^{p-1}y^{q-1}e^{-x-y}\,dx\,dy$ and substitute $x = u^2$, $y = v^2$: it becomes $4\iint_{u,v>0} u^{2p-1}v^{2q-1}e^{-(u^2+v^2)}du\,dv$.

**Hint 2.** In polar coordinates $u = r\cos\theta$, $v = r\sin\theta$ the integral factors: $2\int_0^\infty r^{2(p+q)-1}e^{-r^2}dr = \Gamma(p+q)$ and $2\int_0^{\pi/2}\cos^{2p-1}\theta\sin^{2q-1}\theta\,d\theta = B(p,q)$ (Fact 1). For $p = q = \frac{1}{2}$: $\Gamma(\frac{1}{2})^2 = \Gamma(1)B(\frac{1}{2},\frac{1}{2}) = 2\cdot\frac{\pi}{2} = \pi$.

## Problem 2

**Hint 1.** Theorem 2 is Fact 1 with $2p-1 = n$, $2q-1 = 0$. For even $n = 2m$ use $\Gamma(m+\frac{1}{2}) = \frac{(2m-1)!!}{2^m}\sqrt{\pi}$ to get $\int_0^{\pi/2}\sin^{2m}x\,dx = \frac{(2m-1)!!}{(2m)!!}\cdot\frac{\pi}{2}$; for odd $n = 2m+1$ you get $\frac{(2m)!!}{(2m+1)!!}$.

**Hint 2.** For the product: $I_n = \int_0^{\pi/2}\sin^n x\,dx$ decreases in $n$, so $I_{2m+1} \leqslant I_{2m} \leqslant I_{2m-1}$, and $\frac{I_{2m-1}}{I_{2m+1}} = \frac{2m+1}{2m} \to 1$; hence $\frac{I_{2m}}{I_{2m+1}} \to 1$, which is the Wallis product.

**Hint 3.** $\int_0^{2\pi}\cos^{2020}x\,dx = 4I_{2020} = 2\pi\binom{2020}{1010}2^{-2020}$. For the second, $\sin^2 x\cos^2 x = \frac{1}{4}\sin^2 2x$, so the integral is $\frac{\pi}{8}$.

## Problem 3

**Hint 1.** Bring each integral to the form $\int_0^1 u^{p-1}(1-u)^{q-1}du$: (a) is already $B(\frac{3}{2},\frac{3}{2})$; (b) put $u = x^2$; (c) put $x = t^2$; (d) put $u = x^{2025}$.

**Hint 2.**

(a) $\frac{\Gamma(3/2)^2}{\Gamma(3)} = \frac{\pi}{8}$; (b) $\frac{1}{2}B(\frac{1}{2},\frac{3}{2}) = \frac{\pi}{4}$; (c) $2B(2,\frac{3}{2}) = \frac{8}{15}$; (d) $\frac{1}{2025}B(1,2026) = \frac{1}{2025\cdot 2026}$.

## Problem 4

**Hint 1.** Substitute $x = \frac{t}{1-t}$ (the inverse of $t = \frac{x}{1+x}$): then $1+x = \frac{1}{1-t}$, $dx = \frac{dt}{(1-t)^2}$ and the integral becomes $\int_0^1 t^{a-1}(1-t)^{-a}dt = B(a,1-a)$.

**Hint 2.** $B(a,1-a) = \Gamma(a)\Gamma(1-a) = \frac{\pi}{\sin\pi a}$. The contour method gets the same value from the residue at $-1$ of $\frac{z^{a-1}}{1+z}$ on a keyhole contour; here the work is hidden in the reflection formula (which in turn can be proved by this very integral).

## Problem 5

**Hint 1.** Pointwise, $e^{-x^{2n}} \to 1$ for $|x| < 1$ and $\to 0$ for $|x| > 1$; dominate by $e^{-x^2}$ for $|x| \geqslant 1$ and by 1 for $|x| < 1$. The graph tends to the indicator of $[-1,1]$, a rectangle of area 2.

**Hint 2.** Exactly: $u = x^{2n}$ gives $\int_{-\infty}^{\infty}e^{-x^{2n}}dx = \frac{1}{n}\Gamma\left(\frac{1}{2n}\right) = 2\Gamma\left(1+\frac{1}{2n}\right) \to 2\Gamma(1) = 2$.